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RESEARCH INTERESTS

Low-dimensional topology, in particular braids, knots and links, cobordisms, knot concordance, knotted surfaces, Khovanov and Heegaard Floer homology, 3- and 4-manifolds.

EMPLOYMENT

12/2025 – 11/2027	Postdoctoral fellow , University of Glasgow , Glasgow, UK. Supported by SNSF Postdoc.Mobility fellowship 230329.
10/2023 – 11/2025	Postdoctoral fellow , Max Planck Institute for Mathematics , Bonn, Germany. Mentor: Arunima Ray.
07/2019 – 07/2023	Scientific assistant , ETH Zurich , Zurich, Switzerland.
10/2018 – 04/2019	Teaching assistant , Ulm University , Ulm, Germany.
08/2017 – 05/2018	Teaching assistant , Syracuse University , Syracuse, NY, USA.
10/2013 – 07/2017	Teaching assistant , Ulm University , Ulm, Germany.

EDUCATION

07/2019 – 06/2023	Doctor of Science in Mathematics, ETH Zurich , Zurich, Switzerland. Thesis: <i>On notions of braid positivity and knot concordance</i> . Advisor: Peter Feller.
08/2016 – 02/2019	Master of Science in Mathematics (1.0, with distinction), Ulm University , Ulm, Germany. Minor: Computer Science. Thesis: <i>Massey products and linking numbers</i> . Supervisor: Stefan Wewers.
08/2017 – 05/2018	Master of Science in Mathematics (GPA 4.0), Syracuse University , Syracuse, NY, USA.
10/2012 – 08/2016	Bachelor of Science in Mathematics (1.5), Ulm University , Ulm, Germany. Minor: Physics. Thesis: <i>Das Existenzproblem von Hurwitz</i> . Supervisor: Stefan Wewers.
09/2014 – 02/2015	Visiting semester , ETH Zurich , Zurich, Switzerland.

SELECTED AWARDS

12/2025 – 11/2027	Swiss National Science Foundation Postdoc.Mobility fellowship . <i>Project 230329: Concordance of braid positive knots</i> .
2017	Fulbright Travel Grant .
2014 – 2017	Talanx Scholarship .
2013 – 2014	Deutschlandstipendium Scholarship (Ulm University, EY Foundation).
2012 – 2013	Deutschlandstipendium Scholarship (Ulm University, Rotary Club Ulm).

RESEARCH SEMESTER

Participant in the **AIM Online Research Community *4-Dimensional Topology*** organized by Miriam Kuzbary, Maggie Miller, Juanita Pinzón-Cañedo and Hannah Schwartz in Spring 2021.

PREPRINTS

1. **On the detection of knotted spheres by their traces in high dimensions**, with V. Bais, A. Di Prisa, D. Hartman, C.-S. Hsueh, M. Kegel, A. Merz, M. Pencovitch, A. Ray, D. Santoro and L. Wakelin. [ArXiv:2511.07251](#).

PUBLISHED AND ACCEPTED PAPERS (IN ORDER OF ACCEPTANCE)

8. **Strongly quasipositive links are concordant to infinitely many strongly quasipositive links**. Accepted for publication in *Proc. Amer. Math. Soc.* [ArXiv:2210.06612](#).
7. **Braid positive surgery descriptions**, with M. Kegel.
Accepted for publication in *Glasg. Math. J.* [ArXiv:2512.14589](#).
6. **Non-complex cobordisms between quasipositive knots**, with M. Borodzik.
Accepted for publication in *J. Math. Pures Appl.* (2025), DOI. [ArXiv:2504.04894](#).
5. **Algorithms in 4-manifold topology**, with S. Bastl, R. Burke, R. Chatterjee, S. Dey, A. Durst, S. Friedl, D. Galvin, A. García Rivas, T. Hirsch, C. Hobohm, C.-S. Hsueh, M. Kegel, F. Kern, S. M. S. Lee, C. Löh, N. Manikandan, L. Mousseau, L. Munser, M. Pencovitch, P. Perras, M. Powell, J. P. Quintanilha, L. Schambeck, D. Suchodoll, M. Tancer, A. Thiele, M. Uschold, S. Veselá, M. Weiß and M. von Wunsch-Rolshoven.
Accepted for publication in *Algebr. Geom. Topol.*. [ArXiv:2411.08775](#).
4. **Minimal cobordisms between thin and thick torus knots**, with S. Baader, L. Lewark and F. Misev. *Proc. Edinb. Math. Soc., II. Ser.* 69, no. 1 (2026), pp. 177–183.
3. **On the nonorientable four-ball genus of torus knots**, with F. Binns, S. Kang and J. Simone. *Algebr. Geom. Topol.* 25-4 (2025), pp. 2209–2251.
2. **3-braid knots with maximal 4-genus**, with S. Baader, L. Lewark and F. Misev. *Trans. Amer. Math. Soc. Ser. B* 11 (2024), pp. 600–621.
1. **The upsilon invariant at 1 of 3-braid knots**. *Algebr. Geom. Topol.* 23-8 (2023), pp. 3763–3804.

Preprint versions of these articles are available on [arXiv](#).

CONFERENCE PROCEEDINGS (NOT PEER-REVIEWED)

5. **MATRIX-MFO Tandem Workshop: Invariants in Low-Dimensional Toplogy. Combinatorics, Geometry, and Computation**, with S. Friedl, J. Licata, S. Tillmann. *Oberwolfach Rep.* 21 (2024), no. 3, pp. 2511–2550.
4. **Obstructing ribbon clasp number one**, with D. Celoria, A. Di Prisa, J. Howie, J. Licata, F. Misev, L. Schambeck, L. Wakelin. *Oberwolfach Rep.* 21 (2024), no. 3, pp. 2511–2550.
3. **Notions of braid positivity and knot concordance**. *Oberwolfach Rep.* 21 (2024), no. 3, pp. 2511–2550.
2. **Strongly quasipositive knots are concordant to infinitely many strongly quasipositive knots**. *Oberwolfach Rep.* 20 (2023), no. 1, pp. 172–174.
1. **Concordance of positive braid knots**. *Oberwolfach Rep.* 18 (2021), no. 3, pp. 2360–2362.

INVITED TALKS AT CONFERENCES AND WORKSHOPS

07/2026*	Young Women in Geometric Topology, Bonn.
10/2025	Trisections and related topics, Marseille.
11/2024	IberoSing International Workshop 2024: Low-dimensional Topology & Singularity Theory, Madrid.
07/2024	Mini-symposium <i>Knot theory and low dimensional manifolds</i> , European Congress of Mathematics, Seville.
04/2024	NRW Topology Meeting, Wuppertal.
04/2024	Manifolds and groups in Bologna II, Bologna.
09/2023	Leaning into Topology Workshop, Pisa.
09/2023	Swiss Knots, Regensburg.
06/2023	Mini Workshop, Bern.
03/2023	The low dimensional workshop, Budapest,  .
11/2022	Paroles aux jeunes chercheuses et chercheurs en géométrie et dynamique, Orsay.
09/2022	Mini-symposium <i>Algebra and Low-Dimensional Topology</i> , DMV Annual Meeting 2022, Berlin.
06/2022	Surfaces in 4-manifolds, Le Croisic.

INVITED TALKS AT SEMINARS

02/2026	Geometry and Topology Seminar, Glasgow.
01/2026	Geometry Seminar, MPI MiS Leipzig.
10/2025	Algebra and Geometry Seminar, Neuchâtel.
07/2025	Oberseminar Global Analysis and Geometry, Regensburg.
06/2025	Séminaire Géométrie des Espaces Singuliers, Lille.
05/2025	Geometry seminar, ETH Zurich.
12/2024	Oberseminar Topologie, University of Bonn.
09/2024	Séminaire de Géométrie, Algèbre, Dynamique et Topologie, Dijon.
05/2024	Séminaire de Géométrie et Topologie de Marseille, Marseille.
05/2024	Oberseminar Topologie, Münster.
04/2024	Geometry & Topology Seminar, Glasgow.
03/2024	MPIM Oberseminar, MPIM Bonn.
11/2023	MPIM Topology Seminar, MPIM Bonn, short talk.
05/2023	Low-dimensional topology seminar, MPIM Bonn.
12/2022	Geometry and Topology Seminar, CIRGET, Montréal, online,  .
06/2022	LKS-Seminar, Regensburg.
05/2022	Geometry & Topology Seminar, Caltech, online.
05/2022	Oberseminar Geometrie, Fribourg.
02/2022	Knot Online Seminar (K-OS), online,  .
04/2021	AIM Online Research Community 4-Dimensional Topology, lightning talk.

INVITED TALKS AT GRADUATE COLLOQUIA

12/2023	Pitt AWM Student Seminar, Pittsburgh, online.
10/2023	Geometry Graduate Colloquium, ETH Zurich.
10/2022	Bern–Fribourg Graduate Seminar, Bern.
05/2022	Baby Geometri Seminar, Pisa.
04/2022	Bernoullis Tafelrunde, Basel.
11/2021	Zurich Graduate Colloquium, Zurich.
04/2021	Geometric Topology Grad and Postdoc Seminar (GT GAPS), online.

* means planned.

CONTRIBUTED TALKS AT CONFERENCES AND WORKSHOPS

09/2024	MATRIX-MFO Tandem Workshop: Invariants in Low-Dimensional Topology. Combinatorics, Geometry, and Computation , MFO Oberwolfach, short talk.
02/2024	BTW4: Knot Floer Homology , Besse.
02/2023	Winter Braids XII , Tours.
01/2023	Low dimensional topology , MFO Oberwolfach.
05/2022	Swiss-Mathematical-Society Doctoral Day 2022 , Fribourg.
04/2022	BTW2: Sliceness, exotic pairs, and quantum invariants , Aussois.
09/2021	Invariants and Structures in Low-Dimensional Topology , MFO Oberwolfach, short talk.

OTHER CONFERENCES AND WORKSHOPS ATTENDED

07/2026*	Low-dimensional Topology , MFO Oberwolfach.
06/2026*	Geometry and Topology in Low Dimensions: Lasting Trends and Emerging Directions , Budapest.
12/2025	Workshop on surfaces in 4-manifolds , Regensburg.
10/2025	36. NRW Topology Meeting , Bonn.
07/2025	Categorification in Low Dimensional Topology , Bochum.
06/2025	Swiss Knots , Geneva.
09/2024	Workshop on 4-manifolds and algorithms , Regensburg.
02/2024	Winter Braids XIII , Montpellier.
09/2023	GRAZP: Groups and Rigidity Around the Zimmer Program , Ventotene.
07/2023	Gauge Theory and Topology: in Celebration of Peter Kronheimer's 60th Birthday , Oxford.
01/2023	Winter school on singularities and low-dimensional topology , Budapest.
08/2022	Frontiers in Geometry and Topology Summer School , Trieste.
04/2022	Braids in Low-Dimensional Topology , ICERM, Providence.
08/2021	Perspectives on quantum link homology theories , Universität Regensburg.
07/2021	Tech Topology Summer School , Georgia Institute of Technology, online.
07/2021	Swiss Knots , Fribourg.
06/2021	NCNGT: Nearly Carbon Neutral Geometric Topology Conference , online.
06/2020	NCNGT: Nearly Carbon Neutral Geometric Topology Conference , online.
02/2020	Winter Braids X , Pisa.
10/2019	Regional workshop on foliations and 3-manifolds , Universität Regensburg.
10/2019	Low-dimensional topology workshop , Universität Regensburg.
09/2019	Workshop on 4-manifolds , Max Planck Institute for Mathematics Bonn.
09/2019	Of coarse! Quasi-isometries and groups: rigidity and classification , Ventotene.
07/2019	Swiss Knots , ETH Zurich.
10/2017	27th annual <i>Route 81</i> conference on commutative algebra and algebraic geometry, Cornell University, Ithaca, NY, USA.

SUPERVISION

02/2025 – 10/2025	Master thesis by Chiara Donatone at Università degli Studi di Milano. Thesis title: <i>On RGB links and knot traces</i> . Co-supervision with Diego Matessi. Will continue as PhD student at the University of Glasgow.
06/2024 – 06/2025	Master thesis by Frieda Kern at the University of Bonn. Thesis title: <i>Linking number obstructions to satellite homomorphisms</i> . Co-supervision with Peter Teichner. Continues as PhD student at the University of Glasgow.

TEACHING

	University of Bonn/MPIM Bonn
Winter 2024/25	Lecturer for Topics in 3-manifold topology (Master in Mathematics).
	ETH Zurich
Spring 2022	Course organizer for Analysis II (Bachelor in Civil, Geospatial, or Environmental Engineering).
Fall 2021	Course organizer for Analysis I (Bachelor in Civil, Geospatial, or Environmental Engineering).
Spring 2021	Course organizer for Topologie (Bachelor in Mathematics).
Fall 2020	Course organizer for Lineare Algebra I (Bachelor in Mathematics, Physics, or Interdisciplinary Sciences).
Spring 2020	Course organizer for Algebraic Topology II (Bachelor or Master in Mathematics).
Fall 2019	Teaching assistant for Complex Analysis (Bachelor in Mathematics, Physics, or Interdisciplinary Sciences).
	Syracuse University
Spring 2018	Teaching assistant for Calculus III .
Fall 2017	Teaching assistant for Calculus III .
	Ulm University
Summer 2019	Instructor for Linear Algebra Review Course (Bachelor in Computer Science and Engineering).
Winter 2013/14 – Summer 2019	Student teaching assistant for Analysis I-III and Linear Algebra I-II .

TEACHER TRAINING

Winter 2024/25	Participation in the Teaching Mentoring Program , Hausdorff School for Mathematics in Bonn.
	Mentors: Viktoriya Ozornova and Arunima Ray (both MPIM Bonn).
09/2021	Certificate Learning to Teach .
	Program for doctoral teaching assistants, ETH Zurich.
	Teaching project: <i>How to teach writing proofs—classroom assessment techniques in mathematics classes</i> .

OUTREACH ACTIVITIES

03/2026	Co-organizer of GROW@Dresden 2026 (Graduate Research Opportunities for Women at Dresden 2026), March 26-27, joint with Anita Behme, Türkü Özlüm Çelik, Heather Harrington, Oana Padurariu, Andreas Thom.
03/2025	Invited panelist for panel discussion “ <i>What is mathematics research like?</i> ” at GROW@Frankfurt 2025 .
12/2024	Invited speaker at Tea Time with Women in Mathematics , Bonn.
04/2024	Volunteer at the mentoring lunch at GROW@Bonn 2024 .
03/2023	<i>Work as a PhD Student</i> , invited short talk, MindPhair 2023 , ETH Zurich (job-fair for mathematicians, physicists and computational scientists).
03/2022	<i>Work as a PhD Student</i> , invited short talk, MindPhair 2022 , ETH Zurich.

CONFERENCE AND SEMINAR ORGANIZATION

02/2026 –	ArXiv seminar , University of Glasgow.
10/2023 – 11/2025	Low-dimensional topology seminar , MPIM Bonn.
09/2024	MATRIX-MFO Tandem Workshop: Invariants in Low-Dimensional Topology. Combinatorics, Geometry, and Computation , Mathematisches Forschungsinstitut Oberwolfach, Sept 22–27, joint with Stefan Friedl, Joan Licata and Stephan Tillmann.
09/2020 – 07/2023	Geometry Graduate Colloquium , ETH Zurich, funded by the Zurich Graduate School in Mathematics.

OTHER ACADEMIC SERVICE AND ACTIVITIES

	MPIM Bonn
09/2024	Member of the Postdoc Representative Selection Committee.
	ETH Zurich
09/2021 – 07/2023	Co-organizer of get-togethers for women in math at ETH Zurich and the University of Zurich, funded by the Zurich Graduate School in Mathematics.
08/2021 – 07/2023	Elected member of the Nachdiplom Lectures Committee at the Department of Mathematics.
05/2020 – 05/2023	Member of the Board of VMM (Association of the Mid-level Academic Faculty at the Department of Mathematics).
10/2022	Co-organizer of the first D-MATH (Post)Doctoral Welcome Day.
05/2021 – 05/2022	Mid-level faculty representative in the Department Conference.
	Ulm University
10/2016 – 07/2017	Member of the Mathematics Student Council (Fachschaft Mathematik) at the Department of Mathematics and Economics.

REFEREING AND REVIEWING

2024 – present	Referee and quick opinions for <i>Algebr. Geom. Topol.</i> , <i>Bull. Lond. Math. Soc.</i> and <i>Pac. J. Math.</i>
2024 – present	Reviewer for zbMATH Open .

REFERENCES

Peter Feller , University of Neuchâtel,	peter.feller@unine.ch .
Matthew Hedden , Michigan State University,	mhedden@math.msu.edu .
Lukas Lewark , ETH Zurich,	lukas.lewark@math.ethz.ch .
Viktoriya Ozornova , MPIM Bonn (Teaching),	viktoriya.ozornova@mpim-bonn.mpg.de .
Arunima Ray , University of Melbourne,	aru.ray@unimelb.edu.au .